

Spring Data JPA Query Methods - Implementation Notes

1. Introduction to Query Methods

Spring Data JPA provides a feature called Query Methods that allows developers to create database queries by defining method names inside repository interfaces.

Instead of writing SQL queries manually, Spring Data JPA derives queries automatically from method names.

Example:

```
public interface StudentRepository extends JpaRepository {  
  
    List findByName(String name);  
  
}
```

Spring Data JPA converts this method into an SQL query automatically.

2. Query Method Naming Structure

General format:

find + By + FieldName + Condition

Examples:

findByName()

findByAgeGreaterThan()

findByEmailContaining()

Keywords are used to define filtering conditions.

3. Search by Containing Text

Containing is used when we want to search records where a field contains specific text.

Example:

Repository:

```
List findByNameContaining(String keyword);
```

Generated query concept:

```
SELECT * FROM student
```

WHERE name LIKE '%keyword%';

Use cases:

- Search products by name
- Search users by username
- Search articles by title

4. Filter with Starting Text

StartingWith is used to find records where a column starts with a specific value.

Example:

List findByNameStartingWith(String prefix);

Generated query concept:

SELECT * FROM student

WHERE name LIKE 'prefix%';

Example:

findByNameStartingWith("Van")

Returns:

- Vansh
- Vanita

Does not return:

- Devansh

5. Sorting Query Results

Spring Data JPA supports sorting using OrderBy keyword.

Example:

List findByAgeOrderByNameAsc(Integer age);

Meaning:

- Find students with matching age

- Sort result by name in ascending order

Descending example:

```
List findByAgeOrderByNameDesc(Integer age);
```

6. Fetch Records Between Dates

Between keyword is used to retrieve records between two values.

Example:

```
List findByOrderDateBetween(
```

```
LocalDate startDate,
```

```
LocalDate endDate
```

```
);
```

Generated query concept:

```
SELECT * FROM orders
```

```
WHERE order_date BETWEEN startDate AND endDate;
```

Common usage:

- Date range search
- Reports
- Transactions

7. Greater Than and Lesser Than Queries

GreaterThan:

Example:

```
List findByAgeGreaterThan(int age);
```

Query concept:

```
SELECT * FROM student
```

```
WHERE age > ageValue;
```

GreaterThanEqual:

findByAgeGreaterThanOrEqualTo()

LessThan:

List findByAgeLessThan(int age);

LessThanOrEqualTo:

findByAgeLessThanOrEqualTo()

Used for:

- Price filtering
- Age filtering
- Marks filtering

8. Top Query Methods

Top keyword is used to limit the number of returned records.

Example:

List findTop5ByOrderByMarksDesc();

Meaning:

- Get top 5 students
- Sort by marks descending

Query concept:

SELECT * FROM student

ORDER BY marks DESC

LIMIT 5;

Common usage:

- Top products
- Highest salaries
- Recent records

9. Complete Repository Example

@Entity

```
class Student {
```

```
@Id
```

```
private Long id;
```

```
private String name;
```

```
private Integer age;
```

```
private LocalDate admissionDate;
```

```
}
```

Repository:

```
public interface StudentRepository
```

```
extends JpaRepository {
```

```
List findByNameContaining(String name);
```

```
List findByNameStartingWith(String prefix);
```

```
List findByAgeGreaterThan(Integer age);
```

```
List findByAdmissionDateBetween(
```

```
LocalDate start,
```

```
LocalDate end
```

```
);
```

```
List findTop3ByOrderByAgeDesc();
```

```
}
```

10. Important Query Method Keywords

Containing

- Searches text anywhere in a field

StartingWith

- Searches text from beginning

EndingWith

- Searches text at the end

Between

- Values between two parameters

GreaterThan

- Greater than value

LessThan

- Less than value

OrderBy

- Sort results

Top / First

- Limit results

And

- Combine multiple conditions

Or

- Alternative conditions

11. Advantages of Query Methods

- No need to write SQL for common operations
- Reduces boilerplate code
- Easy to read and maintain
- Automatic query generation
- Works well with Spring Boot applications

12. Summary

Spring Data JPA Query Methods allow developers to create database queries using simple method names.

Flow:

Repository Method

|

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Spring Data JPA

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Hibernate

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Database SQL Query

Query Methods are commonly used for:

- Searching text
- Filtering data
- Sorting records
- Date range queries
- Comparison queries
- Getting top records